LESSON PLAN

BASIC ELECTRICAL (23)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	16
Total No. of Lesson Plans	29	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Wheels, types of wheels and description of typical main wheel
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Wheels, types of wheels and description of typical main wheel
5. Specific Objectives	23.1.3.1 Trainee will be able to describe typical alighting gears of an aircraft and its construction.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Examination of main wheel assembly model; Safety: Handle heavy components with care; Supervision: Instructor monitors handling of simulated brake components
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the aircraft landing gear?
(b) Name two types of wheel configurations used in aircraft.
(c) Why are wheels critical during the alighting phase?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Classification of aircraft wheels • Spoke and disc wheel construction • Hub and axle assembly details • Brake assembly integration • Wheel bearing types and function	L & Ds	PPT, IB	15 min
(b)	• Detailed description of typical main wheel • Construction of wheel rim and tire fit • Function of wheel nut and locking mechanism • Brake disc and caliper arrangement • Shock absorber connection points	L & Ds	PPT, PDF	20 min
(c)	• Comparison of nose wheel vs main wheel • Key construction differences in wheel types • Role of wheel in load distribution • Maintenance implications of wheel design • Summary of alighting gear components	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Describe the construction of a typical main wheel including the hub and rim.
(b) Explain the function of the wheel nut and locking mechanism in the alighting gear.
(c) Differentiate between the construction of a nose wheel and a main wheel.

12. Home Assignment and Follow Up	Prepare a brief note on how the knowledge of wheel construction in this lesson will assist in understanding tyre pressure variations and creep marking covered in LP 17.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Tyres and tubes tyres variation of tyre pressure with all up weight, creep and creep marking on wheels Tyre and brake care	Refer to the master syllabus reference list	LP 17

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min